

NESTING DESIGN · TAKE-HOME

Cut, keep, or scrap

Deciding where every part comes from, and what the metal left behind is really worth.

ACT 1 • THE PROBLEM

Orders come in; we decide how to cut them

Real orders from this batch (jobs.json)



7 parts, 18 × 58"

1" 6061 aluminum · job-2609



5 parts, 22 × 37"

1/4" 7075 aluminum · job-2604

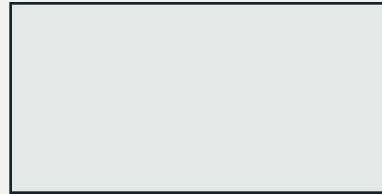


13 parts, 10 × 10"

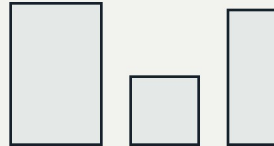
1/2" 6061 aluminum · across 4 orders



Metal we can cut them from



New plate, 60 × 120", bought by weight



Leftovers from earlier jobs (26 on the rack)

For every part, decide:

Which piece of metal?

Where on it?

Turned or not?

What drives the cost, and what I assumed

Metal

\$4.20/lb for 6061, \$7.50/lb for 7075

From the data

Loading

20 minutes per plate or leftover

Brief range → my pick

Machine + labor

\$80/hr, so one load costs about \$27

Brief range → my pick

Cutting

About 1 min per cut; slower if thick or 7075

From the brief

Leftovers

Own orders only. Keeping one: \$7 + \$0.25/sq ft a month

My assumption

Straight cuts

Edge to edge only; parts may turn 90°

From the brief

Saw blade

1/8" of metal lost per cut

Brief range → my pick

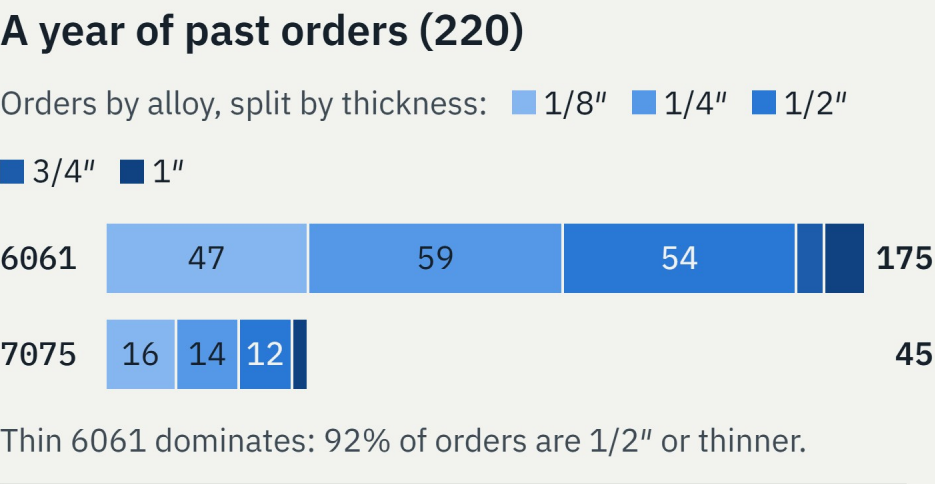
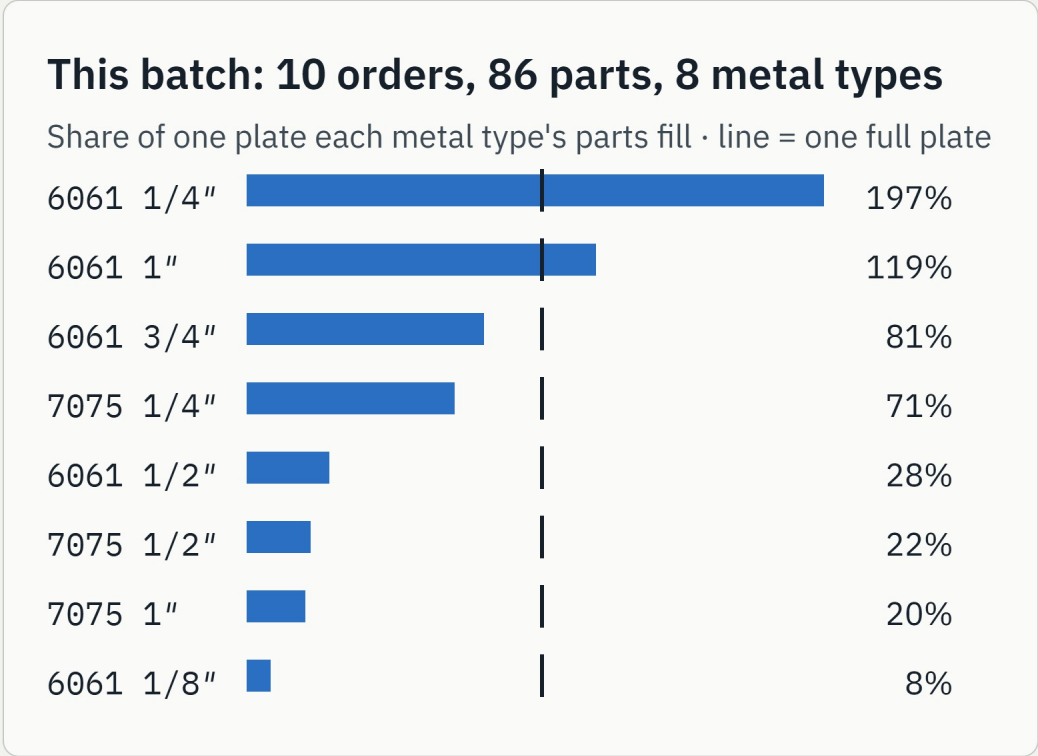
Combining orders

All 10 cut together now; live use: hold ≤ 2 days

My assumption

Tested: a loading cost anywhere from \$20 to \$40 moves the numbers, not the conclusions.

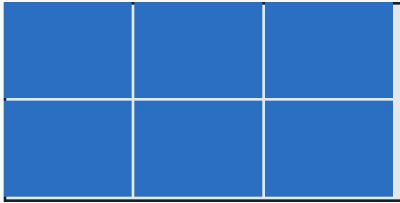
Small batches, and past orders that rarely repeat



88% of part sizes were ordered only once, so the exact same size rarely comes back

Fill plates tightly, use leftovers, then cut the rest

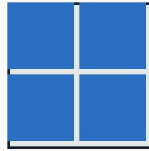
Step 1



Fill whole plates as tightly
as possible



Step 2



Cut the remaining parts
from leftovers on the rack



Step 3

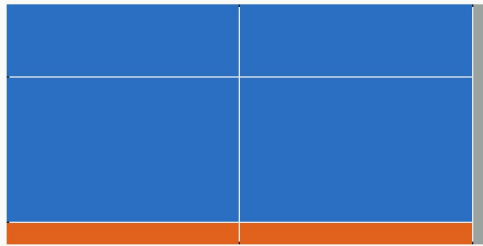


Cut the rest from a
new plate, leaving one
big leftover

Judged by **yield**: the share of the plate that ends up as parts. More is better.

Yield misses what actually costs money

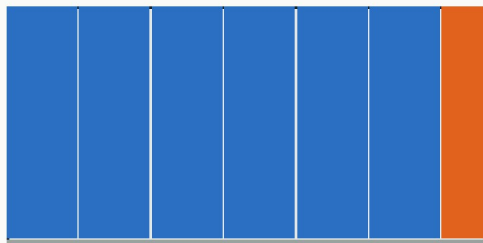
Same yield, different leftovers



A: two thin 5.6" strips

\$32

worth of leftovers



B: one 11 × 58" piece

\$97

worth of leftovers

It ignores loading

Yield loads 8 leftovers here, against 5 in the final plan. Some cost more to load (\$27) than the metal they save.

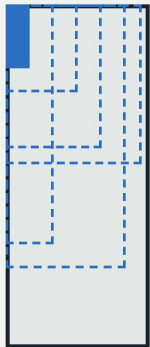
It can't tell useful from useless

A thin strip and a big clean piece both count as the same "waste".

■ Part ■ Leftover ■ Scrap

What is a leftover worth?

Leftover value what keeping it is worth today e.g. \$11.6	=	Chance it gets used from past orders that fit inside it e.g. ~100%	×	Saving if used metal it replaces – \$27 to load it e.g. \$19.2	–	Cost to keep it \$7 to rack + \$0.25/sq ft a month e.g. \$7.6
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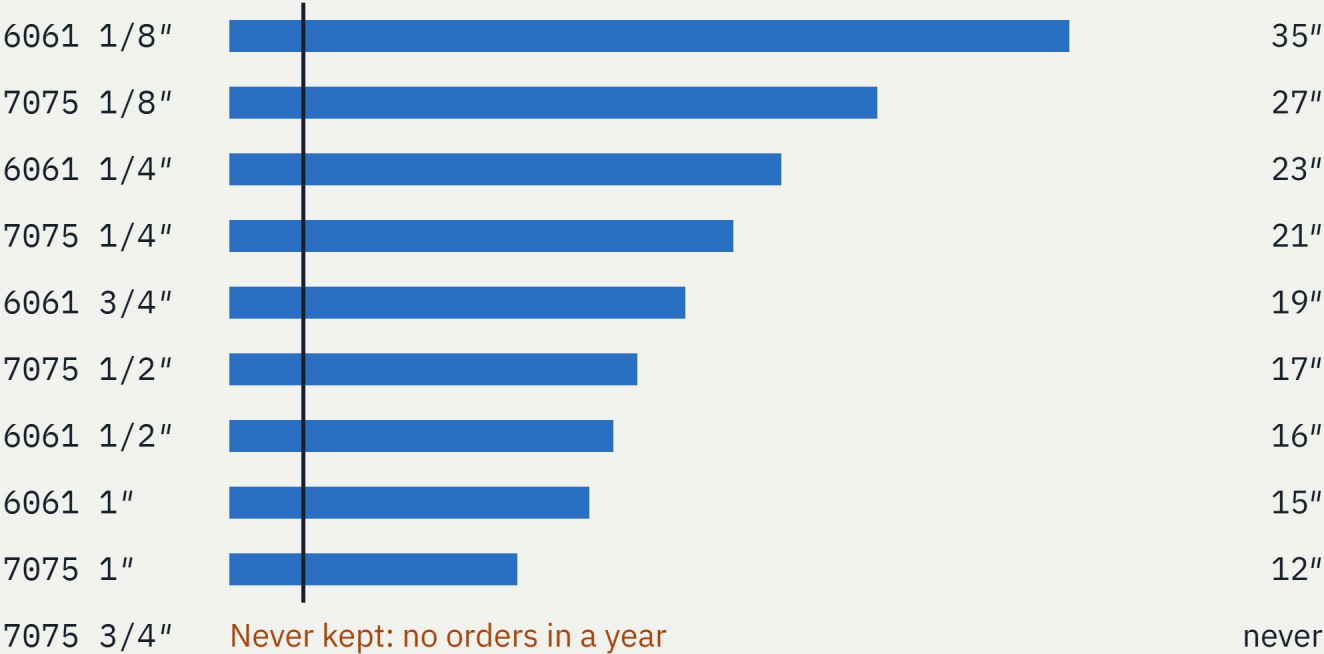
Example: an 18 × 43" leftover

45 of 59 past orders of 1/4" 6061 would have fit inside it (dashed outlines)

It holds **\$79** of metal but is worth only about **\$12**: the \$27 load eats most of the saving.

Most small leftovers are scrap

Smallest leftover worth keeping (side of a square)



Dark line: the smallest part ever ordered (3")

Much bigger than I expected

"Keep anything that fits the smallest part" would keep almost every scrap.

Loading cost decides it

Cheap thin metal needs big leftovers. Expensive thick metal can keep small ones.

No orders, no value

Of the 26 leftovers on the rack today, 15 are worth keeping.

What makes one plan cheaper than another

Pay today

plates bought + loading + cutting

+

Leftovers used up

value spent off the rack

-

Leftovers kept

value put back on the rack

One real decision: thirteen 10 × 10" parts in 1/2" 6061

Way to cut them	Pay today	Used up	Kept	Total
Buy a new plate	\$990	\$0	\$218	\$772
The two biggest leftovers	\$76	\$209	\$33	\$252
Three smaller leftovers	\$104	\$118	\$12	\$210

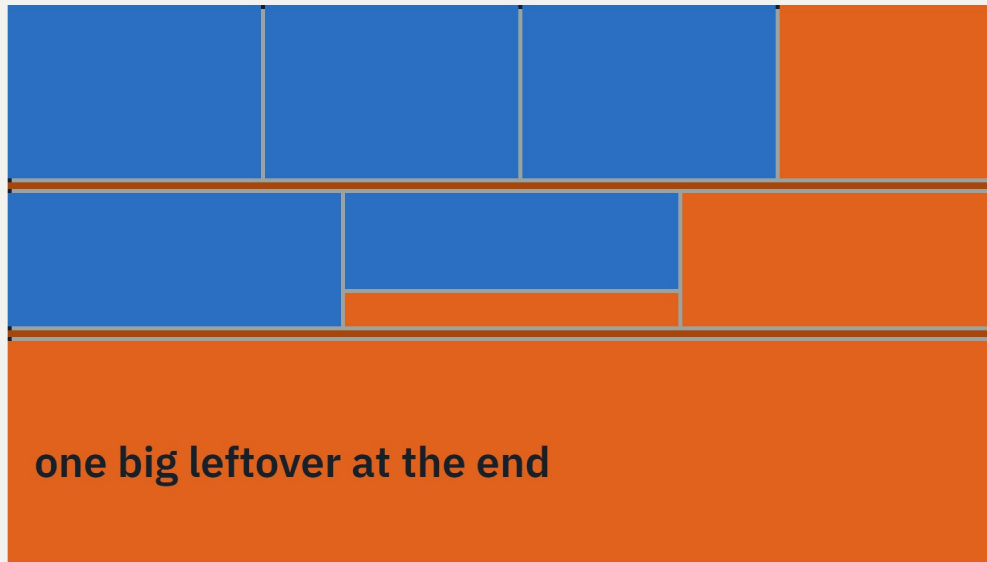
The middle row is **cheapest to cut today** but spends the two pieces future orders are most likely to want. The bottom row pays for one more load and saves them.

Why not just waste least metal?

Wasting least metal would buy the new plate last and grab the biggest leftovers first. It counts no loading cost, and it cannot tell a reusable piece from a sliver.

Only the first block is money we really pay. The other two move value between jobs: credited when a piece is kept, charged at the same value when a later job uses it. So a leftover is never free.

What the saw can actually cut



- **Red lines are the first cuts.** They run edge to edge across the whole plate and make strips.
- **Each strip is then cut into parts.** A strip is as deep as its deepest part.
- **A shorter part gets one trim cut,** which leaves a small piece above it.

Strips can run along either side of the plate, and any part can be turned 90°. The blade eats 1/8" on every cut. Whatever is left at the end stays as one big piece.

This is the constraint every method here has to obey — mine and the computer's. A layout the saw cannot cut is not an option, however tightly it packs.

A quick pass, then a wider search

Step 1 · the quick pass

Fill the best piece, then repeat



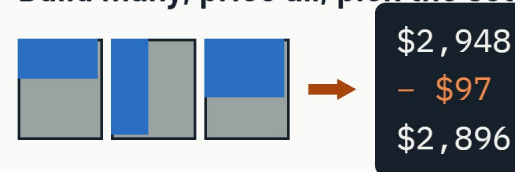
Take the fullest layout for the parts still unplaced. Cut it. Repeat with what is left, until every part has a home.

Fast and easy to explain. But it commits to each piece before it has seen the rest.

On this batch the two passes mostly agree. For 1/4" 6061 the quick pass already lands on the final plan. Over all 8 metals the search saves **\$672**, almost all of it in better leftovers rather than less metal bought.

Step 2 · the search

Build many, price all, pick the set



Build every sensible layout for every plate and leftover. Price each one with the cost rule: pay today, plus leftovers used, minus leftovers kept. Then pick the cheapest **combination** that cuts all the parts.

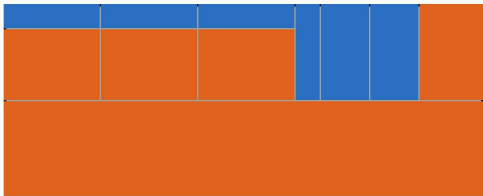
45 layouts built for the 1/4" 6061 order; 4 chosen. Under 5 seconds per metal.

Known exact methods exist (arc-flow, branch-and-price) that prove the best answer for classic cutting problems. They need a clean linear objective; our leftover value is a lookup on the piece's size, which breaks them. So this is the heuristic version of the same decomposition.

What the search adds

Same plate, same 8 parts, better leftovers

1" 7075 • eight 6 × 24" parts on one 48 × 120" plate • drawn to scale



Quick pass only

Mixes both orientations in one strip. Leaves four awkward pieces and one big one.

\$3,801



After the search

All parts the same way round. Leaves two clean rectangles worth \$870.

\$3,561

Same metal bought, same cuts, \$240 apart. The whole difference is what is left behind.

Two loads that never paid

3/4" 6061 • five 10 × 10" parts



13×38



11×24

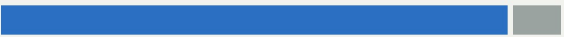
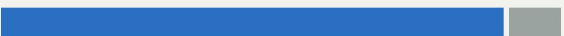
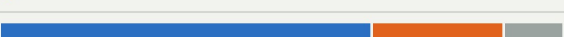
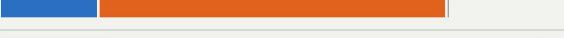
The quick pass pulls these two off the rack for five small parts.

Two extra loads at \$27, plus \$29 of leftover value spent — for parts that fit on the plates already open.

The search skips both. **\$238 cheaper.**

■ Part ■ Kept ■ Scrap

Where the metal went, piece by piece

Metal	Piece cut	Bar length = size of the piece	Parts
6061 1/8"	Plate 48 x 96		6
6061 1/4"	Leftover 17 x 21		1
6061 1/4"	Plate 48 x 120		13
6061 1/4"	Plate 48 x 120		14
6061 1/4"	Leftover 25 x 36		2
6061 1/2"	Leftover 29 x 45		8
6061 1/2"	Leftover 22 x 22		4
6061 1/2"	Leftover 10 x 15		1
6061 3/4"	Plate 48 x 120		10
6061 3/4"	Plate 48 x 120		3
6061 1"	Plate 60 x 120		6
6061 1"	Plate 60 x 120		3
7075 1/4"	Plate 48 x 120		5
7075 1/2"	Plate 48 x 96		2
7075 1"	Plate 48 x 120		8

 **53%**

became parts

 **40%**

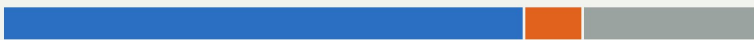
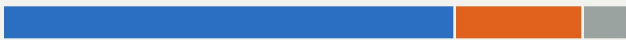
went back on the rack as leftovers worth keeping

 **7%**

scrap, including saw cuts

5 of the 15 pieces were leftovers from the rack, not new plates.

A year of real orders, replayed

Metal	Plate metal bought: parts · still on rack · scrap	Plates	Leftovers used
6061 1/8"		46	43
6061 1/2"		38	62
6061 1/4"		25	45
7075 1/8"		15	14
7075 1/4"		11	17
6061 1"		6	8
7075 1/2"		9	13
6061 3/4"		6	4
7075 1"		2	2

 **73%** became parts

 **14%** still on the rack

 **13%** scrap

158 new plates and **208 leftovers** used for 220 order lines.

Same year, quick pass only
161 plates, scrap 15.4%, and **\$2,480 more cash**

With a bigger budget, I would attack loading next



Automate loading

Robots fetch the right piece from the rack and load the saw.

20 → 5 min

The smallest leftover worth keeping shrinks by a third or more (1/2" 6061: 16" → 10").



Cut any shape

A laser or waterjet cuts any path, so parts nest tighter.

Bigger leftovers

Waterjet suits thick aluminum better than laser; test on real parts.



Plan smarter

Decide per order: cut now, or wait for a better match.

Learn from data

Simulate future orders; learn real cut times from machine logs.

First version (a few weeks): the optimizer, a monthly leftover-value table, and a log of every leftover.